Marking scheme: Microeconomics exam

University, BSc Economics, year 1

Total: 40 points

General instructions for the marker

- For each criterion, award full points, partial points (in steps of 0.5) or 0 points, as set out in the table.
- An error carried forward into a later sub-question is penalised only once. If the student correctly continues with their own wrong result, they earn the points for the next step.
- Different wording is fine as long as the economics is correct.
- In a fill-in table each column is one criterion. All cells in a column correct: full points. One cell wrong: half. Two or more wrong: 0.
- Grade = 1 + 9 × (points earned / total points), rounded to one decimal.

Question 1. Opportunity cost and sunk cost (3 points)

On Saturday, Sanne can work 8 hours in a café for €14 per hour, or study for this exam. She has paid €2,530 in tuition fees and bought a textbook for €60.

a. Calculate the opportunity cost of spending Saturday studying.

b. Explain why the tuition fees and the textbook should play no role in her decision for Saturday.

Context for the marker

This question tests opportunity cost (the best forgone alternative) and sunk cost. The tuition fees and the book have already been paid and do not change with the choice for Saturday, so they are not relevant to the decision.

Model answer

a. The best forgone alternative is working: 8 × €14 = **€112**.

b. The tuition fees and the book are **sunk costs**: they have already been incurred and cannot be recovered, whatever Sanne chooses on Saturday. Only costs and benefits that change with the decision are relevant.

Marking criteria

Criterion	Max.	Full points if	Partial points
Best forgone alternative	0.5	The student identifies working as the best forgone alternative.	No half points.
Calculation	1	€112.	0.5 points: tuition or textbook included, or a calculation error.
Concept of sunk cost	0.5	The concept of sunk cost is named or described.	No half points.
Why it is irrelevant	1	Already paid and unaffected by the choice, so not relevant to the decision.	0.5 points: only ‘already paid’, without saying it does not change with the choice.

Question 2. Shifts and movements (3 points)

Consider the market for e-bikes.

a. Cheaper batteries cause the price of e-bikes to fall. Is this a movement along the demand curve or a shift of the demand curve? Explain.

b. The price of petrol rises sharply. What happens to the demand curve for e-bikes? Explain.

Context for the marker

This question tests the difference between a shift of a curve and a movement along a curve. Cheaper batteries shift supply; along the demand curve the quantity demanded rises. Higher petrol prices shift the demand for e-bikes (a substitute for the car).

Model answer

a. Lower production costs: the **supply curve shifts to the right**. The price falls and there is a **movement along the demand curve**: quantity demanded rises.

b. The **demand curve shifts to the right**: the e-bike is a substitute for the car, which becomes more expensive to use.

Marking criteria

Criterion	Max.	Full points if	Partial points
a. Cause on the supply side	1	Supply shifts right due to lower costs.	0.5 points: lower costs mentioned without a supply shift.
a. Movement along the demand curve	1	Movement along the demand curve, quantity demanded rises.	0.5 points: ‘more demand’ without distinguishing demand from quantity demanded.
b. Shift in demand	1	Demand shifts right, with the substitute relationship.	0.5 points: correct direction without a reason.

Question 3. Equilibrium and surplus (5 points)

In the market for bicycle lights (Q in thousands of units, P in euros):

$Q_d = 200 - 4P$

$Q_s = -40 + 2P$

a. Calculate the equilibrium price and quantity.

b. Calculate consumer surplus and producer surplus at the equilibrium.

Context for the marker

This question tests equilibrium and calculating surplus as triangles. The demand curve meets the price axis at P = 50, the supply curve at P = 20. Consumer surplus = ½ × (50 − 40) × 40; producer surplus = ½ × (40 − 20) × 40. Amounts in thousands of euros.

Model answer

a. 200 − 4P = −40 + 2P → **P = 40**, **Q = 40** (40,000 units).

b. Demand meets the price axis at P = 50, supply at P = 20.

Consumer surplus = ½ × (50 − 40) × 40 = **200**, i.e. €200,000.

Producer surplus = ½ × (40 − 20) × 40 = **400**, i.e. €400,000.

Marking criteria

Criterion	Max.	Full points if	Partial points
Equilibrium price	1	P = 40.	0.5 points: correct method, calculation error.
Equilibrium quantity	1	Q = 40, or correctly carried forward from the student's own wrong price.	0.5 points: calculation error.
Consumer surplus	1.5	200 (× €1,000) with the correct triangle.	1 point: correct triangle with a calculation error. 0.5 points: wrong height of the triangle (for example the equilibrium price itself instead of 50 − 40).
Producer surplus	1.5	400 (× €1,000) with the correct triangle.	1 point: correct triangle with a calculation error. 0.5 points: wrong height of the triangle.

Question 4. Excise tax and tax incidence (5 points)

In the market from question 3, the government levies an excise tax of €6 per unit, paid by the producer.

a. Calculate the new equilibrium.

b. Calculate the tax revenue and the deadweight loss.

c. Who bears most of the tax, consumers or producers? Explain using elasticities.

Context for the marker

This question tests tax incidence. New supply function Qs = −52 + 2P. Consumers pay €2 more, producers receive €4 less. The side with the less elastic curve bears more: here supply is relatively inelastic (steeper) compared with demand.

Model answer

a. Qs = −40 + 2(P − 6) = −52 + 2P. 200 − 4P = −52 + 2P → **P = 42**, **Q = 32**.

b. Revenue = 6 × 32 = **192** (€192,000). Deadweight loss = ½ × 6 × (40 − 32) = **24** (€24,000).

c. Consumers pay €2 more (42 − 40), producers receive €4 less (40 − 36). **Producers bear most**, because supply is **less price elastic** than demand: producers are less able to adjust.

Marking criteria

Criterion	Max.	Full points if	Partial points
New equilibrium	1.5	P = 42 and Q = 32.	1 point: one calculation error. 0.5 points: tax applied in the wrong direction.
Tax revenue	1	192 (× €1,000), or correctly carried forward.	0.5 points: unit or calculation error.
Deadweight loss	1	24 (× €1,000).	0.5 points: correct triangle without ½, or a calculation error.
Division of the burden	0.5	Consumers €2, producers €4, so producers bear most.	No half points.
Explanation with elasticity	1	The less elastic side bears more; here that is supply.	0.5 points: elasticity mentioned but the wrong side identified, or no link to the division.

Question 5. Point elasticity (4 points)

Take the equilibrium from question 3 (P = 40, Q = 40).

a. Calculate the price elasticity of demand at this point.

b. Is demand elastic or inelastic here, and what happens to total consumer spending if the price rises slightly?

Context for the marker

This question tests point elasticity E = (dQ/dP) × (P/Q) = −4 × 40/40 = −4. With elastic demand, total spending (revenue) falls when the price rises.

Model answer

a. E = (dQ/dP) × (P/Q) = −4 × 40/40 = **−4**.

b. |E| > 1: demand is **elastic**. When the price rises, quantity falls proportionally more than the price rises, so **total spending falls**.

Marking criteria

Criterion	Max.	Full points if	Partial points
Formula	1	Point elasticity using dQ/dP × P/Q.	0.5 points: formula inverted, or arc elasticity instead of a point.
Result	1	−4.	0.5 points: calculation error, or +4 without the sign but correctly interpreted.
Elastic	1	Elastic, because /E/ > 1.	0.5 points: correct conclusion without the criterion.
Total spending	1	Falls, with an explanation.	0.5 points: correct direction without explanation.

Question 6. Costs (4 points)

De Korenaar bakery has fixed costs of €12,000 per month, variable costs of €1.50 per loaf and a selling price of €3.50.

a. Calculate the break-even quantity.

b. At 8,000 loaves per month, calculate the profit and the average total cost.

Context for the marker

This question tests break-even and average total cost. An error carried forward is penalised only once.

Model answer

a. €12,000 / (€3.50 − €1.50) = **6,000 loaves**.

b. Profit = 8,000 × €2.00 − €12,000 = **€4,000**. ATC = (€12,000 + 8,000 × €1.50) / 8,000 = **€3.00**.

Marking criteria

Criterion	Max.	Full points if	Partial points
Break-even	1.5	6,000.	1 point: calculation error. 0.5 points: fixed costs divided by the price.
Profit	1	€4,000.	0.5 points: one type of cost forgotten.
Average total cost	1.5	€3.00.	1 point: correct formula with a calculation error. 0.5 points: only average fixed or average variable cost.

Question 7. Monopoly and deadweight loss (5 points)

A firm holds a patent on a medical test and is the only supplier. The inverse demand function is P = 100 − 2Q. Marginal cost is constant: MC = 20. There are no fixed costs.

a. Determine the profit-maximising quantity and price.

b. Calculate the profit.

c. Calculate the deadweight loss compared with perfect competition.

Context for the marker

This question tests MR = MC without MR being given: the student must derive MR = 100 − 4Q. Under perfect competition P = MC = 20, so Q = 40. The deadweight loss is the triangle ½ × (60 − 20) × (40 − 20) = 400.

Model answer

a. Revenue = (100 − 2Q)Q, so MR = 100 − 4Q. MR = MC → **Q = 20**, **P = 60**.

b. Profit = (60 − 20) × 20 = **800**.

c. Under perfect competition: P = MC = 20 → Q = 40. Deadweight loss = ½ × (60 − 20) × (40 − 20) = **400**.

Marking criteria

Criterion	Max.	Full points if	Partial points
Deriving MR	1	MR = 100 − 4Q.	0.5 points: MR derived incorrectly but the MR = MC rule applied.
Quantity and price	1	Q = 20 and P = 60.	0.5 points: only Q correct, or P read from the MR curve.
Profit	1	800.	0.5 points: calculation error, or revenue instead of profit.
Quantity under perfect competition	0.5	Q = 40.	No half points.
Deadweight loss	1.5	400 with the correct triangle.	1 point: correct triangle with a calculation error. 0.5 points: rectangle instead of triangle (½ forgotten) or wrong corners.

Question 8. Externalities (3 points)

A chemical plant discharges waste water into a river, so fishermen downstream catch less.

a. Using marginal costs, explain why the market leads to an inefficient outcome.

b. Describe a Pigouvian tax: how high should it be, and why does it restore efficiency?

Context for the marker

This question tests negative externalities using marginal analysis. Marginal social cost lies above marginal private cost, so the market produces more than the social optimum. A Pigouvian tax equal to marginal external cost at the optimum internalises the externality.

Model answer

a. **Marginal social cost** (private cost plus the damage to fishermen) lies above **marginal private cost**. The plant produces where price equals its private cost, so **more than the social optimum**.

b. A per-unit tax **equal to the marginal external cost** at the social optimum. The private cost curve then coincides with the social cost curve: the plant pays for the damage (internalising it) and chooses the efficient quantity by itself.

Marking criteria

Criterion	Max.	Full points if	Partial points
Social versus private cost	1	MSC lies above MPC.	0.5 points: damage mentioned without marginal costs.
Overproduction	0.5	The market produces more than the social optimum.	No half points.
Level of the tax	0.5	Equal to the marginal external cost.	No half points.
Mechanism	1	Internalisation: private cost becomes social cost, the market chooses the optimum.	0.5 points: ‘the plant will discharge less’ without the mechanism.

Question 9. Game theory (3 points)

Two airlines each choose a high or low price. Profits in millions of euros (A ; B):

	B high	**B low**
A high	8 ; 8	2 ; 10
A low	10 ; 2	4 ; 4

a. Does A have a dominant strategy? Determine the Nash equilibrium.

b. Explain why this game is a prisoner's dilemma.

Context for the marker

This question tests dominant strategies and Nash equilibrium. Low is dominant for both (10 > 8 and 4 > 2), the Nash equilibrium is (low, low), while (high, high) is better for both.

Model answer

a. Yes: **low is dominant** for A (10 > 8 if B chooses high, 4 > 2 if B chooses low), and by symmetry for B too. Nash equilibrium: **(low, low)** with 4 ; 4.

b. Both high (8 ; 8) is better for both, but each firm has an incentive to deviate to low. Individually rational behaviour leads to a collectively worse outcome: that is the **prisoner's dilemma**.

Marking criteria

Criterion	**Max.**	**Full points if**	**Partial points**
Dominant strategy	1	Low is dominant, with a comparison of the payoffs.	0.5 points: correct strategy without comparison.
Nash equilibrium	1	(low, low).	0.5 points: correct equilibrium without justification.
Prisoner's dilemma	1	Cooperation is better, but the incentive to deviate makes it unstable.	0.5 points: only 'both high is better'.

Question 10. Market structures (fill-in table) (5 points)

Complete the table. The table has **5 columns**: number of firms, product (homogeneous or differentiated), barriers to entry, pricing (relation between P and MC) and long-run economic profit. The **4 rows** are the four market structures.

Market structure	**Number of firms**	**Product**	**Barriers to entry**	**Pricing**	**Long-run economic profit**
Perfect competition					
Monopolistic competition					
Oligopoly					
Monopoly					

Context for the marker

This is a fill-in table. Mark per column: each column is one criterion. A column with all 4 cells correct earns 1 point, 3 correct cells earn 0.5 points, 2 or fewer earn 0. Pay particular attention to the last column: under monopolistic competition long-run economic profit is zero because of free entry.

Model answer

Market structure	Number of firms	Product	Barriers to entry	Pricing	Long-run economic profit
Perfect competition	Very many	Homogeneous	None	Price taker, P = MC	Zero
Monopolistic competition	Many	Differentiated	Low	Limited market power, P > MC	Zero (free entry)
Oligopoly	A few	Homogeneous or differentiated	High	Strategic, depends on rivals, P > MC	Possibly positive
Monopoly	One	Unique, no close substitutes	Very high or blocked	Price maker, MR = MC, P > MC	Possibly positive

Marking criteria

Criterion	Max.	Full points if	Partial points
Column number of firms	1	All 4 cells correct.	0.5 points: 3 of 4 correct.
Column product	1	All 4 cells correct; for oligopoly both answers are correct.	0.5 points: 3 of 4 correct.
Column barriers to entry	1	All 4 cells correct.	0.5 points: 3 of 4 correct.
Column pricing	1	All 4 cells correct.	0.5 points: 3 of 4 correct.
Column long-run economic profit	1	All 4 cells correct.	0.5 points: 3 of 4 correct.